Week 4: Class Exercise

Q1

A string with a linear density of 525 g/m is subjected to a 50N tension. A sinusodial wave with a frequency of 100Hz and an amplitude of 8.5mm is sent along the string. Determine the average rate of wave energy transported along the string.

Solution.

i/ Calculate the velocity of the wave.

$$v = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{50N}{0.525\ kg/m}} = 9.76\ m/s$$

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ii/ Average rate of energy transport (wave) = Average Power of the wave

$$P_{ave} = \frac{1}{2}\mu v \omega^2 y_m^2$$

$\omega = 2\pi f$

$$= \frac{1}{2}(0.525)(9.76)(2\pi \times 100)^2 (0.0085)^2$$

$$\approx 73.08\ W$$

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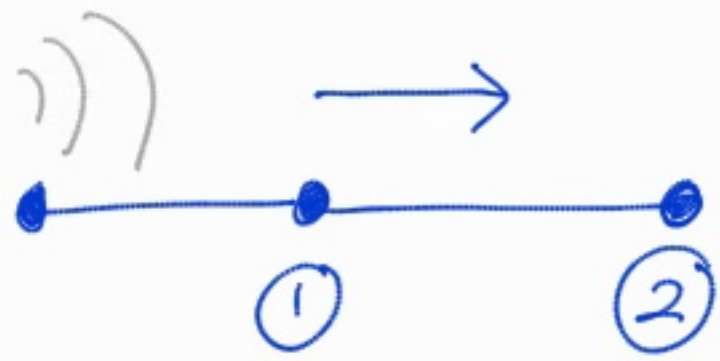
Consider a scenario where you are overhearing a very heated conversation that registers 80 dB. Thus you move some distance away where the sound intensity reduces by a factor of 200. Determine the sound intensity level at your new distance away from that conservation.

(3)

Solution.

$$\beta = (10\,dB)\log_{10}\left(\frac{I}{I_0}\right)$$

$$I_2 = \frac{I_1}{200}$$



$$\beta_1 = (10\,dB)\log_{10}\left(\frac{I_1}{I_0}\right) = 80\,dB$$

$$\beta_2 = (10\,dB)\log_{10}\left(\frac{I_2}{I_0}\right)$$

$$\beta_2 - \beta_1 = (10\,dB)\log_{10}\left(\frac{I_2}{I_1}\right)$$

$$= (10\,dB)\log_{10}\left(\frac{\cancel{I_2}}{200\cancel{I_2}}\right)$$

$$= (-2.301)(10\,dB)$$

$$\beta_2 = -23.01\,dB + 80\,dB$$

$$\approx 57\,dB \;\#$$